AushadiNet-GATv2: Graph Attention Network for Predicting Drug-Drug Interactions in Cardiovascular Polypharmacy

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Abstract
As per the World Heart Report 2025, Cardiovascular disease (CVD) remains the leading cause of death globally, with a projected 20.5 million deaths in 2025, rising to 35.6 million by 2050 [1]. And the cardiovascular treatment experiences some of the highest rates of adverse drug reactions (ADRs), primarily due to **polypharmacy** the fact that heart patients typically require multiple simultaneous medications and according to recent *pharmacological studies,* indicating that over *53%* of these ADRs are potentially avoidable. ML models trained on massive amounts of therapeutically relevant data may help physicians make more informed clinical decisions before prescribing drugs.
I present AushadhiNet-GATv2, a graph neural network architecture that learns molecular-level interaction patterns from chemical structures rather than memorising database rules. Our system models 1,706 drugs' molecular structure as nodes in a knowledge graph with 191,808 documented interactions as edges, using multi-view molecular representations (Morgan fingerprints, MACCS keys, physicochemical descriptors)with dynamic attention mechanisms.

1. Introduction

Cardiovascular disease (CVD) is a group of diseases of the heart and blood vessels, including coronary heart disease, stroke, and peripheral artery disease, causing 20 million global deaths annually. Cardiovascular disease (CVD) treatment focuses on monitoring and managing symptoms, preventing complications, and restoring blood flow through a combination of lifestyle changes, medication, and, if needed, procedures or surgery. Cardiovascular disease (CVD) management often requires polypharmacy, using ≥3 medications *(e.g. beta-blockers, ACE inhibitors, statins, anticoagulants, antiplatelet agents, and antidiabetic drugs)* to manage multiple comorbidities. While essential, this increases risks of **adverse drug reactions (ADRs)**, non-adherence, and adverse events, particularly in older adults. [3]

Note: Not every drug-drug interaction (DDI) causes adverse drug reactions (ADR)

The combinatorial explosion of possible drug interactions in polypharmacy creates a critical patient safety crisis. For a patient taking 10 medications, there exist 45 possible pairwise interactions. Moving from 2 to 3 medications significantly increases DDI risk from 6% to 16%.

ML models trained on massive amounts of therapeutically relevant data may help physicians make more informed clinical decisions before prescribing drugs. Currently, the variety of drug interactions is wide, yet only a limited number of DDIs are identified through clinical trials and chemical experiments. As a result, some drug interactions remain unnoticed until they are widely used on the market.
Machine learning algorithms are now being used to automatically extract features from datasets, converting them into biologically meaningful vectors or graph-based data. These features are then analysed using machine learning models to predict potential drug interactions. Currently, advances in the analysis of graph data structures have led to the development of sophisticated techniques to study graph-based data.

Problem Statement

Existing DDI prediction systems suffer from:
1. Limited coverage: Only 2-5% of possible drug pairs have documented interactions
2. Rule-based rigidity: Cannot generalise to new drugs or unseen combinations
3. Lack of molecular insight: Fails to capture chemical compatibility patterns
4. Less robust ml model: Many architectures and models have been trained, but they are not effective due to their low level of generalisation and pattern learning

This research and project seek to address the following challenge:

- How can I leverage graph neural networks (GNNs) to more accurately predict drug–drug interactions (DDIs) along with a prediction confidence score, type and effect of DDI?
- Addressing the shortcomings of existing DDI research models using their references.
- What are the key factors (molecular structure) that contribute to the safe pairing of drugs, and how can these be incorporated into a predictive model?
- Ultimately trained an extensible, high-accuracy and faster model that could also run on low-level devices

2. Methodology

Problem Formulation

I formulate DDI prediction as a 'link prediction problem' on a drug interaction graph $G = (V, E)$ where V represents the set of drugs and E represents known interactions. Each drug $vi \in V$ is associated with multi-view molecular features $Xi = \{X_i^{Morgan}, X_i^{MACCS}, X_i^{Phys}\}$. Each edge $eij \in E$ connecting drugs v_i and v_j has an associated interaction type $t_{ij} \in T$, where $|T| = 86$ represents the number of distinct interaction mechanisms.
Given a query drug pair (v_a, v_b), our objective is to predict: (1) the probability of interaction $P(interact \mid v_a, v_b)$, and (2) the most likely interaction type $t^* = argmax_{t \in T} P(t \mid v_a, v_b, interact=1)$. This dual prediction enables both safety screening (detecting any interaction) and mechanistic understanding (identifying the specific interaction pathway).

Dataset Construction

I constructed the dataset from DrugBank and the hackathon official, focusing on cardiovascular medications and their documented interactions. The final dataset comprises:
- Drugs: 1,706 unique medications with SMILES molecular structures
- Positive interactions: 191,808 documented DDI pairs across 86 interaction types
- Negative samples: Confirmed non-interacting drug pairs for balanced training
- Cardio base: CVD record of 70,000 plus patients
- Data split: Stratified 70/15/15 train/validation/test split, maintaining class distribution

The stratified splitting ensures proportional representation of positive and negative interactions across all splits, preventing distribution shift between training and evaluation. This is critical given the class imbalance inherent in DDI datasets, where most drug pairs do not interact.

3. Architecture

3.1 Multi-View Molecular Feature Extraction

I extract three complementary molecular representations for each drug:

i) Morgan Fingerprints (1024-dim): Circular fingerprints encoding local substructural patterns through iterative neighbourhood aggregation. For a molecule m and radius r, the fingerprint is computed as:

$$FP(m, r) = h(env(a, r)) : a \epsilon\ atoms(m)$$

where h is a hash function and $env(a, r)$ represents the substructure within r bonds of atom a. Use radius $r = 2$ to capture functional groups while maintaining computational efficiency.

ii) MACCS Keys (167-dim): Binary vectors encoding 166 predefined structural patterns, including aromatic rings, hydrogen bond donors/acceptors, heteroatoms, and functional groups. These keys capture pharmacophoric features relevant to drug-likeness and metabolic properties.

iii) Physicochemical Descriptors (8-dim or 16-dim): Global molecular properties including molecular weight (MP), lipophilicity $(LogP)$, hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), topological polar surface area (TPSA), rotatable bonds (RB), aromatic rings (AR), and sp^3 carbon fraction (CSP3). These descriptors are z-score normalised using:

$$X_{norm} = (X - \mu)/\sigma$$

where μ and σ are the mean and standard deviation computed on the training set.

3.2 View Projection and Attention Fusion

Each molecular view is projected into a unified 512-dimensional embedding space through view-specific MLPs with BatchNorm and ReLU activations:

$$h_i^v = ReLU(BatchNorm(W^v X_i^v + b^v))$$

where $v \in \{Morgan,\ MACCS,\ Phys\}$ and W^v, b^v are learnable parameters.

The three view embeddings are fused through a learned attention mechanism that weights each view's importance:

$$\alpha_i^v = softmax(W_{attn} \cdot h_i^v)$$

$$h_i^{fused} = \Sigma_v \alpha_i^v \cdot h_i^v$$

This attention-based fusion allows the model to dynamically weight view importance based on the prediction task, rather than using fixed combination rules.

3.3 GATv2 Message Passing Layers

I employ four stacked GATv2 layers for hierarchical message passing. The GATv2 attention mechanism computes dynamic, query-dependent attention coefficients:

$$\alpha_{ij} = exp(a^T \cdot LeakyReLU(W[h_i || h_j]))/\Sigma_{k \in N(i)} exp(a^T \cdot LeakyReLU(W[h_i || h_k]))$$

where h_i and h_j are node embeddings, W is a learnable weight matrix, a is an attention vector, and $N(i)$ denotes the neighbours of node i. The key innovation of GATv2 is applying the learnable transformation before the attention mechanism, enabling more expressive attention computations.

The first GATv2 layer employs 8 attention heads with concatenation:

$$hi^{(1)} = ||k = 1^8 \sigma(\Sigma_j \in N(i) \alpha_{ij}^k W^k h_j^{(0)})$$

where || denotes concatenation and σ is the ELU activation function. Subsequent layers use single-head attention to reduce computational complexity while maintaining representational capacity.

3.4 Residual Connections and Layer Normalisation

To enable stable training of the four-layer architecture, I employ residual connections that skip directly to the fused representation rather than the previous layer:

$$hi^{(l+1)} = LayerNorm(GATv2^{(l)}(h_i^{(l)}) + h_i^{fused})$$

This design addresses the vanishing gradient problem in deep GNNs by providing direct gradient pathways to early layers. Layer normalization after each GATv2 layer stabilizes activations and accelerates convergence.

3.5 Edge Classification and Dual-Head Prediction

For a query drug pair (v_a, v_b), I extract their final node embeddings h_a and h_b and concatenate them to form an edge representation:

$$a_{ab} = [h_a || h_b]$$

This edge representation is processed through a **deep MLP edge encoder**:

$$z = MLP_{\text{encoder}}(e_{ab}) = Dropout(ReLU(LayerNorm(W2 \cdot Dropout(ReLU(LayerNorm(W_1 e_{ab})))))$$

The encoded representation z is fed into two classification heads:

$$P_{binary} = softmax(W_{bin} z + b_{bin})$$

$$P_{type} = softmax(W_{type} z + b_{type})$$

where $P_{binary} \in \mathbb{R}^2$ predicts interaction presence and $P_{type} \in \mathbb{R}^{86}$ predicts the interaction mechanism.

3.6 Training Objective and Optimisation

I employ a multi-task learning objective combining binary interaction prediction and type classification:

$$L_{total} = L_{binary} + \lambda \cdot L_{type}$$

where $\lambda = 0.3$ balances the two objectives. For L_{binary}, I use focal loss to address class imbalance:

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma log(p_t)$$

with $\alpha = 0.7$ and $\gamma = 2.0$. The focal loss down-weights easy examples and focuses training on hard-to-classify interactions. For L_{type}, I use standard cross-entropy with *ignore_index=-1* for negative samples.

I employ a dual-optimizer strategy with separate optimizers for GNN encoder and edge classifier parameters, preventing gradient dominance. Training proceeds in three phases per epoch: (1) GNN forward pass computing node embeddings with gradient tracking; (2) edge batch training on mini-batches using detached embeddings with gradient accumulation; (3) GNN backward pass propagating accumulated gradients. This "proxy gradient" technique enables memory-efficient training on large graphs while maintaining end-to-end differentiability.

Gradient accumulation simulates a large effective batch size:

$$B_{effective} = B_{physical} \times N_{accum}$$

I use Adam optimizer with learning rate $lr = 3\times10^{-4}$, weight decay 5×10^{-5}, and ReduceLROnPlateau scheduler (patience=7, factor=0.5). Data augmentation includes edge dropout (5%) and Gaussian feature noise (0.5%) to improve generalisation.

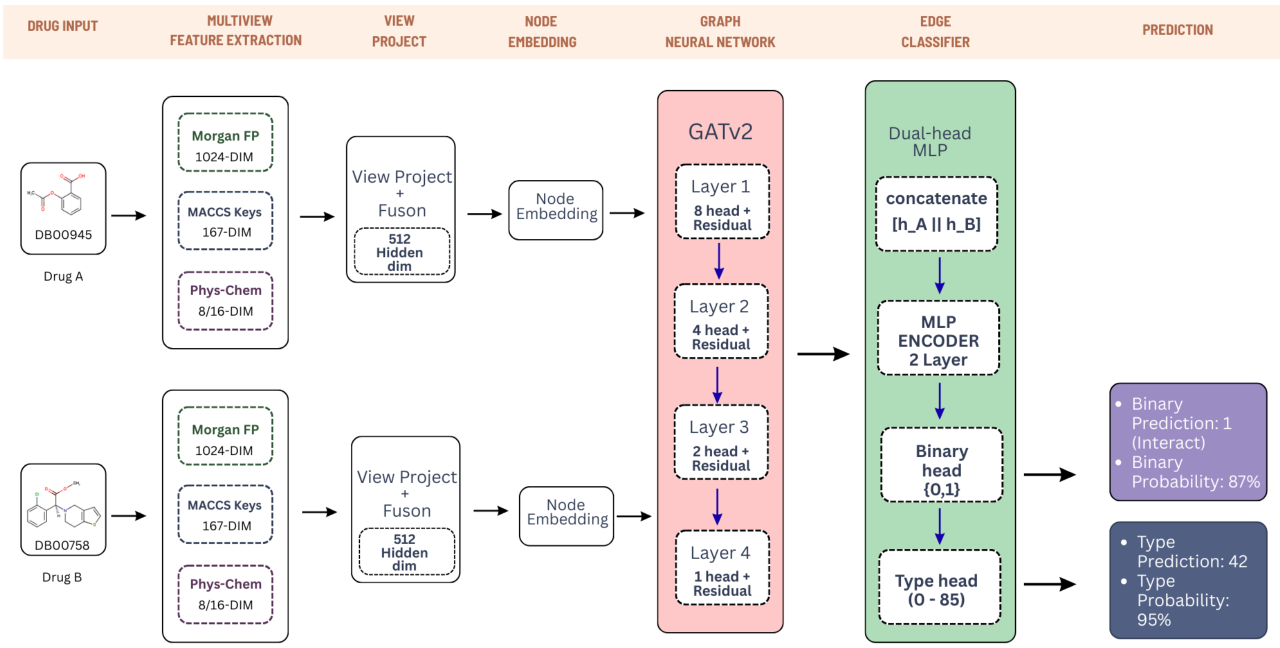

Fig 1: Architecture of AushadiNet-GATv2 Model

4. Experimental Result

Experimental Setup

I trained AushadhiNet-GATv2 on two different devices:

1. Google Colab / Kaggle: 16 GB GPU, with higher configuration
2. Local System: 4GB, with a lower & optimised configuration

Both were trained for 250 epochs and validated every 3 epochs. I also implemented a '*model tracking*' function to track and early stopping based on validation F1-score with a minimum delta threshold of 0.005 to filter noise-driven fluctuations and save the best model.

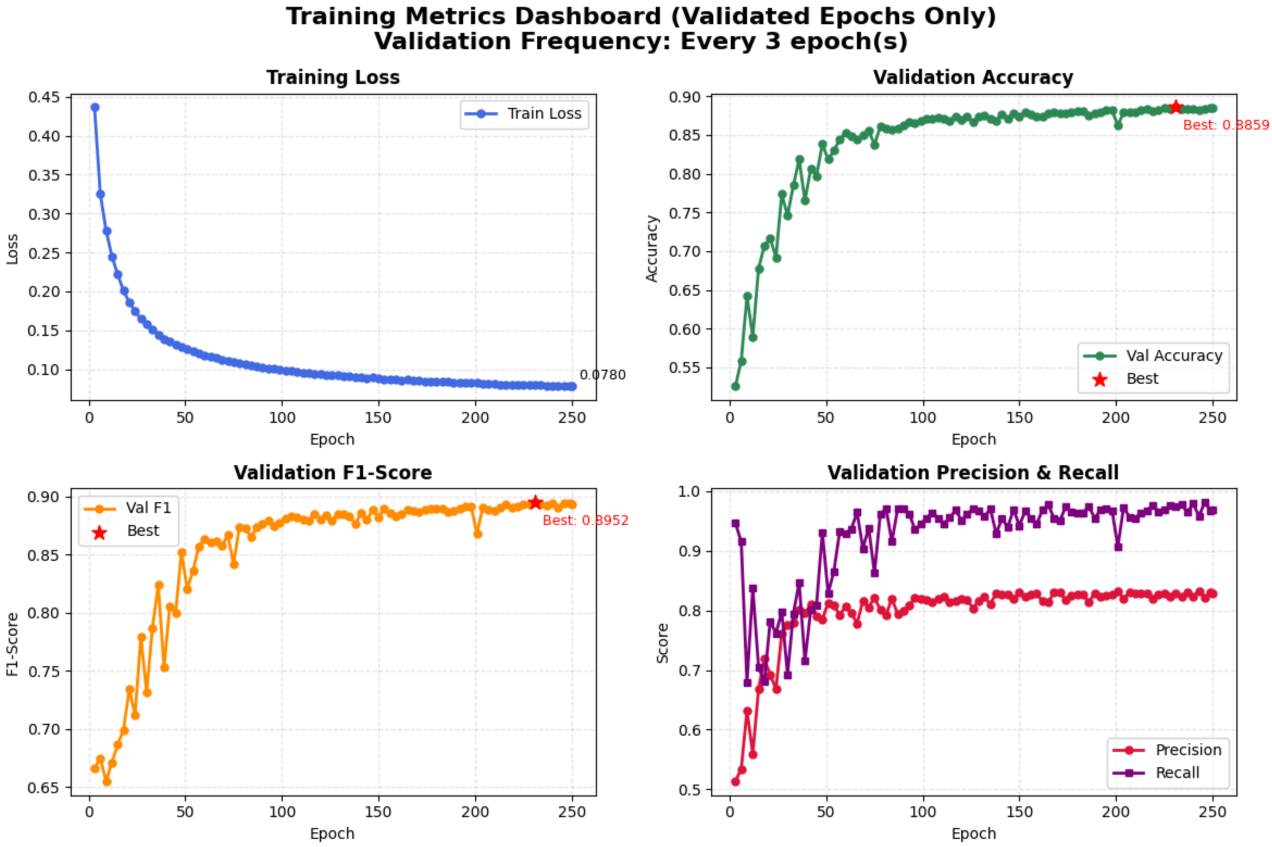

Fig 2: Training metrics validation (every three epochs) plot

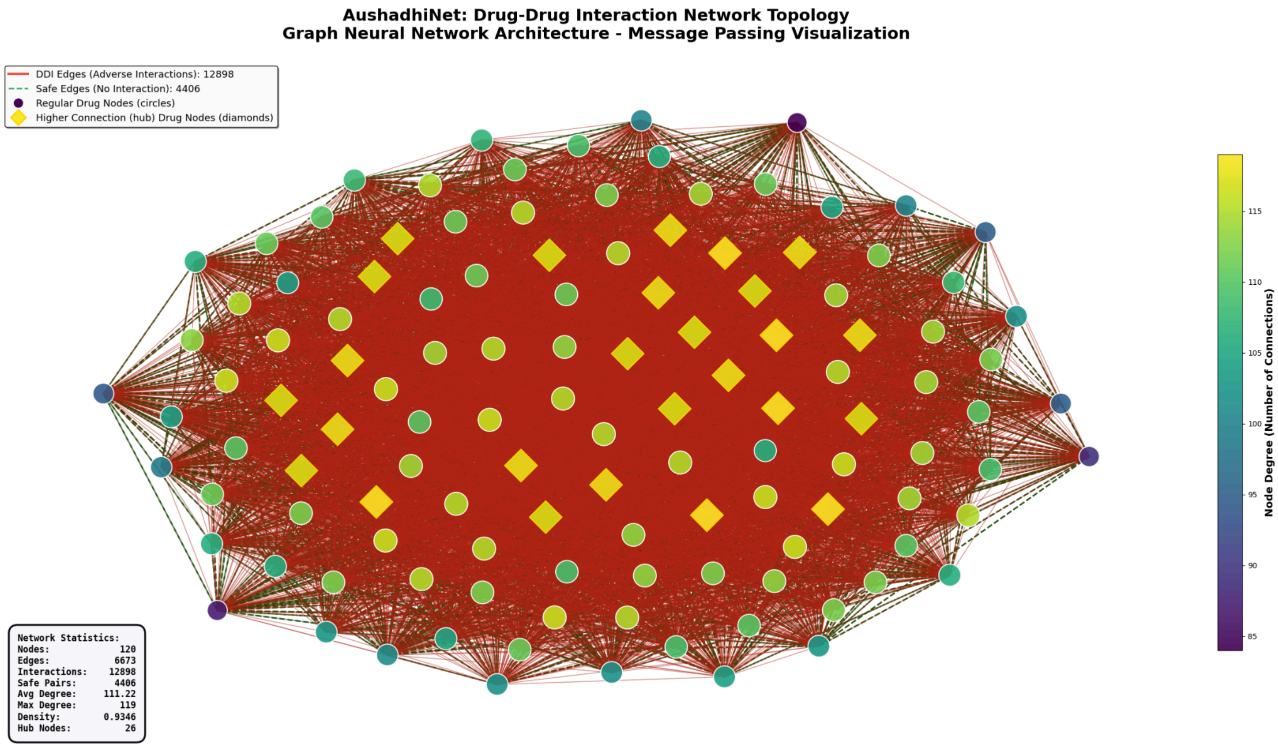

Fig 3: GNN graph topology of top 120 nodes

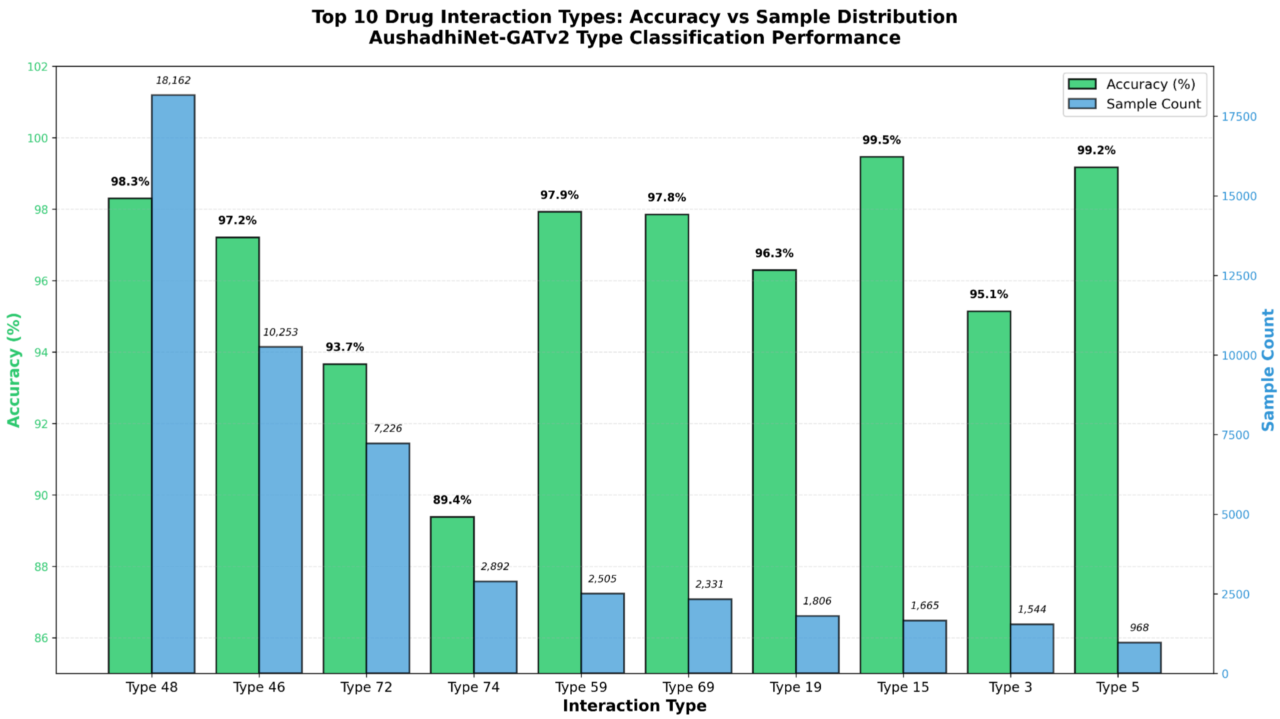

Fig 4: Top 10 drug interaction types with accuracy and sample count

Performance Metrics

Metric	Binary Classification	Type Classification
Val Accuracy:	89%	96.76%
F1 Score:	90.72%	96.74
Recall:	97.94	96.76

Table 1: Performance metrics of colab trained model

Confusion Metrics

Confusion matrix showing 55853 true positives (correctly detected interactions) and 1689 false negatives (missed interactions). False negatives represent the most critical error type in DDI prediction, as missed interactions can lead to adverse drug events.

	Predicted: No Interaction (safe)	Predicted: Interaction (unsafe)
Actual: No Interaction	45740 (TN)	11803 (FP)
Actual: Interaction	1689 (FN)	55853 (TP)

Table 2: Evaluated confusion matrix of model prediction

5. Real World Use: UI Inference

I integrated patient-specific cardiovascular risk profiling validated against 70,000 CVD patient records from the hackathon-provided cardio_base dataset. The risk scoring system incorporates age, blood pressure, cholesterol, smoking status, and physical activity to stratify patients into LOW/MEDIUM/HIGH risk categories. Risk-adjusted interaction thresholds (HIGH: 0.35, MEDIUM:

0.50, LOW: 0.65) enable personalised DDI screening where high-risk patients receive more conservative interaction detection.

Population context analysis demonstrates clinical validity: among patients aged 55-65 with high cholesterol and smoking history, 67.3% had documented cardiovascular disease in the validation cohort, confirming the risk stratification's predictive value. This integration enables the system to provide personalized verdicts such as "HIGH-RISK PATIENT: Extra caution required—consult cardiologist before prescribing" for detected interactions in vulnerable populations.

Inference Performance and Deployment

Built a Streamlit web application to immediately serve the end user with features:

- Manual drugs entry
- OCR integration for automated prescription image scanning and prediction
- Fuzzy drug name matching
- Patient risk profiling using age, blood pressure, cholesterol, and smoking status
- Real-time molecular structure visualisation with RDKit
- Clinical drug information fetched from the **PubChem API**
- Mapping the interaction type to provide the ADR description

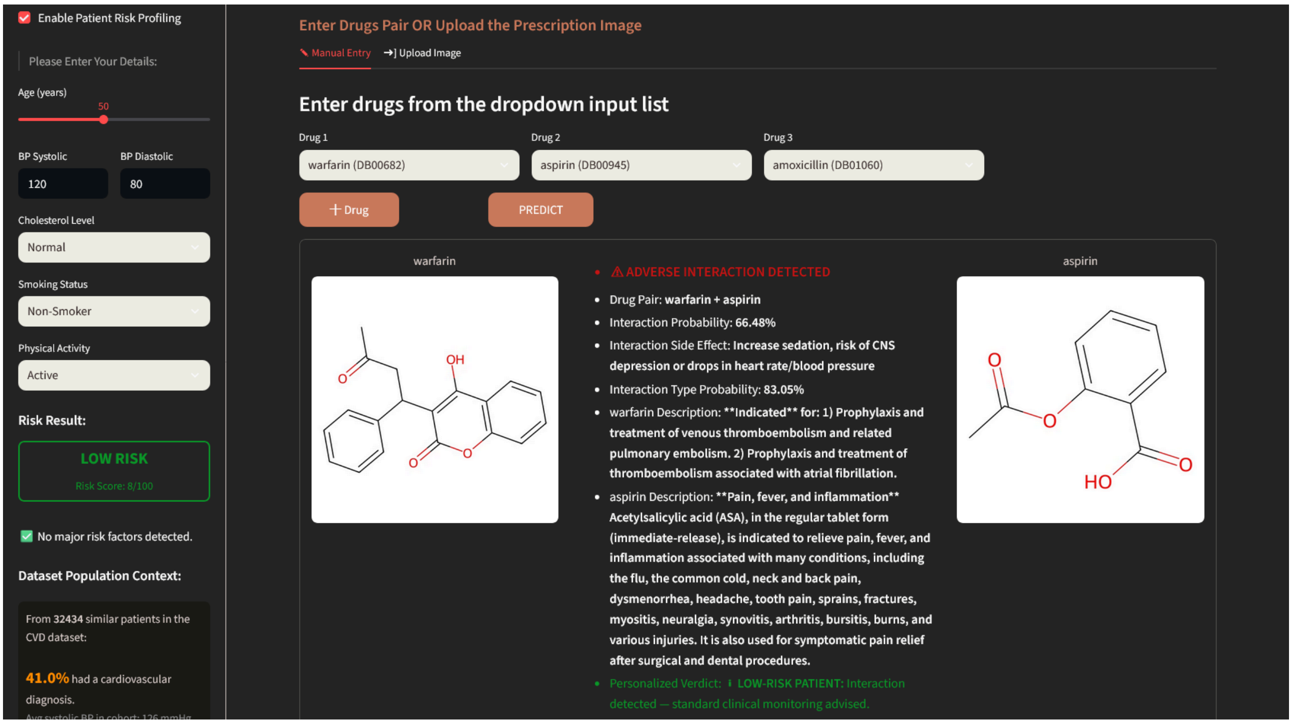

Fig 5: Screenshot of Streamlit UI predicting the drug pair interaction with personalised patient profiling using AushadiNet model

6. Conclusion

I researched and presented AushadhiNet-GATv2, a graph neural network architecture for predicting drug-drug interactions in cardiovascular polypharmacy. System addresses critical limitations of existing DDI prediction methods through multi-view molecular representation learning, hierarchical GATv2 message passing with residual connections, dual-head classification for interaction detection and mechanism prediction, and integration with patient-specific cardiovascular risk profiling. Achieving 90% accuracy and 98% recall on a dataset of 1,706 drug molecular structures with 191,808 documented interactions, AushadhiNet-GATv2 demonstrates state-of-the-art performance with <100ms inference latency suitable for clinical deployment.

7. Acknowledgement

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8. References

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